Maxwell Clarke

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EDUCATION

University at Buffalo – SUNY Buffalo, NY
BS Electrical Engineering | GPA 3.0 **August 2024**
MS Electrical Engineering **Expected May 2026**

PROFESSIONAL EXPERIENCE

Aaron Clarke Consulting May 2024 – Present

Role: Machine Learning Engineer

- Authored blog post tutorials on ML deployment for AI development boards
- Developed software for an embedded system using LiteRT
- Programmed microcontrollers to run lightweight AI inference tasks
- Developed Bluetooth module using PSoC Creator
- Performed Yocto builds and created hardware images

Elysium Security (Startup) September 2024 –Present

Role: Co-Founder

- Analyzed Gateway devices to be used for relaying camera streams to cloud storage
- Developed a Python-Based UDP socket system
- Compared accuracy for various gesture recognition ML algorithms using PyTorch
- Conducted license plate recognition using Yolov8 and EasyOCR

Creekside Energy Solutions (Startup) September 2024 – Present

Role: Co-Founder

- Analyzed electrical components for use in a small-scale hydroelectric turbine
- Won a pitch competition held by the UB Startup and Innovation Collaboratory
- Drafted Structure for AI-based solar forecasting website

RESEARCH EXPERIENCE

University at Buffalo Nanosatellite Laboratory April 2025 – October 2025

Communication Systems Team Lead

- Utilized link budget to select antennas for communication with ground station
- Researched FCC standards for radio communication to small satellites
- Authored slides and presented on the COMs subsystem in a review
- Led weekly meetings held for researching satellite components
- Authored Trade Study on Frequency Bands for Satellite Communications

TECHNICAL SKILLS

- C/C++
- Python
- Pytorch/MNIST
- VHDL
- Yocto
- Embedded Linux
- Qiskit
- Vitis/Vivado
- LiteRT